

Hackspectra

Problem Brief: Smart Parking Management System

In crowded cities, finding parking is time-consuming and leads to traffic congestion and fuel waste. The goal is to develop a smart system that uses sensors to detect available parking spaces and provides real-time information to users via a mobile app, allowing them to locate and reserve parking efficiently.

Product Requirement Document (PRD)

Smart AI-Based Parking Management System

1. Product Overview

The Smart Parking Management System is an AI-powered solution designed to reduce urban parking congestion by enabling users to locate, reserve, and manage parking slots efficiently.

The system integrates:

-  User Interface (for booking & tracking)
-  Admin Dashboard (for monitoring & control)
-  AI Camera System (for automation & detection)

It supports both **manual and fully automated modes**, ensuring flexibility and scalability.

2. Objectives

- Reduce traffic congestion caused by parking search
- Optimize parking space utilization
- Automate parking operations using AI
- Provide real-time parking availability

- Enable seamless user experience
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3. Stakeholders

- End Users (Drivers)
 - Parking Operators / Admin
 - System Developers
 - Smart City Authorities (future scope)
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4. Key Features

4.1 User Interface (User App)

Functionalities:

- View nearby parking stations
 - Check real-time slot availability
 - Book parking slots
 - View assigned slot number
 - Track parking duration (live timer)
 - View pricing details
 - Receive automated updates
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4.2 Admin Dashboard

Functionalities:

- Monitor all parking slots
- View real-time occupancy
- Manually assign/reassign slots

- Toggle automation mode
 - View analytics:
 - Vehicle type distribution
 - Peak usage hours
 - Revenue tracking
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4.3 AI Camera System

Functionalities:

- Detect vehicle (car/bike)
- Recognize number plate (ANPR)
- Track entry and exit time
- Send data to backend

Technologies:

- OpenCV
 - YOLO (object detection)
 - OCR (EasyOCR/Tesseract)
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4.4 Smart Slot Allocation

Description:

Dynamic allocation of parking slots based on vehicle type.

Logic:

- If bike count > car count → convert car slots into multiple bike slots
 - If car demand increases → prioritize larger slots
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4.5 Automated Pricing System

Functionalities:

- Track parking duration automatically
 - Calculate cost based on time
 - Display price in user interface
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4.6 Automation Mode Toggle (Core Feature)

Description:

A switch that allows system operation in either manual or automated mode.

Modes:



Manual Mode:

- Admin manages slot allocation
 - Entry/exit handled manually
 - Pricing controlled manually
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Automated Mode:

- Camera detects vehicle
 - System assigns slot automatically
 - Entry/exit tracked via AI
 - Pricing calculated automatically
 - No human intervention required
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UI Behavior:

- Toggle switch labeled **"Automation Mode"**
- Status indicator:
 - ON → Automated

- OFF → Manual
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5. System Flow

Automated Mode Flow:

1. Vehicle enters parking
 2. Camera detects vehicle type & number
 3. Slot assigned dynamically
 4. Timer starts
 5. Vehicle exits
 6. Duration calculated
 7. Price generated
 8. User notified
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6. System Architecture

Components:

- Frontend (User + Admin UI)
 - Backend Server
 - Database
 - AI Detection Module
 - Camera Input System
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7. Technology Stack

Layer	Technology
Frontend	React / Flutter
Backend	Node.js / Firebase

Layer	Technology
Database	Firebase / MongoDB
AI/ML	OpenCV, YOLO, OCR
Hardware	Camera / IoT Sensors
Maps	Google Maps API

8. Data Requirements

- User details
 - Booking records
 - Slot availability status
 - Vehicle type & number plate
 - Entry & exit timestamps
 - Pricing data
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9. Constraints & Assumptions

Constraints:

- Camera accuracy may vary
- Real-time processing limitations
- Hardware dependency

Assumptions:

- Internet connectivity available
 - Cameras installed at entry/exit
 - Limited parking slots (for MVP)
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10. MVP Scope (Hackathon Version)

Must Build:

- User UI with booking
- Admin dashboard
- Slot management system
- Automation toggle
- Simulated AI detection

Optional:

- Real camera integration
- Advanced analytics